ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 2 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable sketches.

SECTION A

QUESTION 1

1.1 1.1.1 C ✓ 1.1.2 A ✓ 1.1.3 B ✓ 1.1.4 D ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 D ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 J ✓ 1.3.2 A ✓ 1.3.3 I ✓ 1.3.4 E ✓ 1.3.5 H ✓ 1.3.6 G ✓ 1.3.7 C ✓ 1.3.8 F ✓ 1.3.9 D ✓ 1.3.10 B ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 (a) $V_m = 325$ V ✓ (b) $V_{rms} = 0{,}707 \times 325$ ✓ $= 229{,}78$ V ✓ (c) $\omega = 2\pi f \rightarrow f = 314 \div (2 \times 3{,}142)$ ✓ $= 49{,}97$ Hz (≈ 50 Hz) ✓ (d) $T = 1 \div f = 0{,}02$ s ✓ (6)

2.2.1 $\tau = R \times C$ ✓ $= 10\,000 \times 100 \times 10^{-6} = 1$ s ✓ (2)

2.2.2 5τ ✓ $= 5$ s ✓ (2)

2.2.3 Curve rising quickly then levelling off towards 9 V ✓; 63% (5,67 V) marked at τ = 1 s ✓; almost fully charged at 5τ, axes labelled ✓ (3)

2.3 $X_L = 2\pi fL$ ✓ $= 2 \times 3{,}142 \times 50 \times 0{,}2$ ✓ $= 62{,}84\ \Omega$ ✓ (3)

[16]

QUESTION 3

3.1.1 $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$ ✓ $= \sqrt{(30^2 + 40^2)}$ ✓ $= 50\ \Omega$ ✓ (3)

3.1.2 $I = V \div Z$ ✓ $= 230 \div 50 = 4{,}6$ A ✓ (2)

3.1.3 Inductive ✓ – X_L is larger than X_C, so the current lags the voltage ✓ (2)

3.2 $f_r = 1 \div (2\pi\sqrt{(LC)})$ ✓ $= 1 \div (2 \times 3{,}142 \times \sqrt{(0{,}01 \times 1 \times 10^{-6})})$ ✓ $= 1 \div (6{,}284 \times 1 \times 10^{-4})$ ✓ $= 1\,591{,}34$ Hz ✓ (4)

3.3 Below resonance the circuit is capacitive and the impedance is high; as resonance is reached the impedance drops to its minimum (= R) ✓ and the current rises to its maximum ✓; above resonance the circuit becomes inductive and the impedance rises again, so the current falls ✓ (3)

[14]

QUESTION 4

4.1 Diode symbol with a gate lead from the cathode side ✓; anode ✓, cathode ✓ and gate ✓ labelled. Switched on by a positive pulse on the gate while the anode is positive with respect to the cathode ✓; switched off only when the anode current falls below the holding current (or the supply is removed / reversed) ✓ (6)

4.2 Any TWO ✓✓: light dimmers; motor speed control; controlled rectifiers / battery chargers; crowbar over-voltage protection; heater / temperature control (2)

4.3.1 $f = 1{,}44 \div ((1\,000 + 2 \times 10\,000) \times 10 \times 10^{-6})$ ✓ $= 1{,}44 \div 0{,}21$ ✓ $= 6{,}86$ Hz ✓ (3)

4.3.2 Any ONE ✓: flashing warning LED; clock pulses for a counter; beeper / buzzer; simple PWM / stepper pulses (1)

4.4 Sinusoidal oscillator produces a sine-wave output ✓, e.g. RC phase-shift, Wien bridge, LC (Colpitts / Hartley), crystal ✓; non-sinusoidal oscillator produces square, triangular or sawtooth waves ✓, e.g. 555 astable, multivibrator, relaxation oscillator ✓ (4)

[16]

QUESTION 5

5.1 (a) $L_T = 4 + 12$ ✓ $= 16$ mH ✓ (b) $L_T = (4 \times 12) \div (4 + 12)$ ✓ $= 3$ mH ✓ (4)

5.2 A slotted / patterned disc on the motor shaft passes between a light source and a sensor (or a magnet past a Hall sensor) ✓; each slot produces a pulse ✓; the controller counts pulses per second to calculate speed ✓; channels A and B are 90° out of phase (quadrature) ✓; whichever channel leads shows the direction of rotation ✓ (5)

5.3.1 $n_s = 120 \times 50 \div 4$ ✓ $= 1\,500$ r/min ✓ (2)

5.3.2 $\text{slip} = (n_s - n_r) \div n_s \times 100$ ✓ $= (1\,500 - 1\,440) \div 1\,500 \times 100$ ✓ $= 4\%$ ✓ (3)

5.4 The three-phase currents in the stator windings produce a magnetic field that rotates at synchronous speed ✓; the field cuts the rotor conductors ✓ and induces currents in them ✓; these currents create their own magnetic field that interacts with the stator field ✓, producing a torque that turns the rotor in the direction of the rotating field (always slightly slower – slip) ✓ (5)

5.5 Any TWO ✓✓: simple and rugged construction; little maintenance (no brushes); cheap; reliable; almost constant speed; easily controlled with a VFD (2)

5.6 The VFD rectifies the AC supply to DC and then inverts it back to AC at an adjustable frequency ✓; synchronous speed $n_s = 120f \div p$ depends on frequency ✓, so changing the frequency changes the motor speed (the voltage is adjusted with it) ✓ (3)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE